ITMaths past paper - Jan 2018.

Sample Solutions.

Q1 (Compulsory)

a) $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 2 & -2 \\ 1 & 3 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 6 & 0 \\ -3 & 1 & 10 \end{bmatrix}$ $C = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$

i) size - Rank A = 3×3
Rank B = 2×3
Rank C = 3×1

size and rank mean the same thing here. given by: Rows × cols.

ii) elements $b_{22} = 1$ $C_{21} = 3$

second Row, second column

iii) Transpose - Rows become cols and cols become rows.

$C^T = [1 \ 3 \ 4]$

b) $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & -5 \\ 1 & 1 & -2 \end{bmatrix}$

i) cofactors C_{11}, C_{12} and C_{13}. (expand across Row 1, Find submatrices ...)

a_{11} $\begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & -5 \\ 1 & 1 & -2 \end{bmatrix}$ submatrix

$\begin{bmatrix} 1 & -5 \\ 1 & -2 \end{bmatrix}$ ad-bc

$1(-2) - (-5)(1) = -2 - (-5) = 3$

$M_{11} = 3$

$\therefore C_{11} = 3$

a_{12} $\begin{bmatrix} 1 & 2 \\ 3 & -5 \\ 1 & -2 \end{bmatrix}$ submatrix $\begin{bmatrix} 3 & -5 \\ 1 & -2 \end{bmatrix}$ $ad-bc$

$3(-2)-(-5)(1) = -6-(-5) = -1$

$M_{12} = -1$ $\quad i+j = 1+2 = (3)$ odd

$\therefore C_{12} = -M_{12} = -(-1) = 1$

a_{13} $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & -5 \\ 1 & 1 & -2 \end{bmatrix}$ submatrix $\begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$ $ad-bc$

$3(1)-(1)(1) = 3-1 = 2$

$M_{13} = 2$ $\quad i+j = 1+3 = (4)$ even

$\therefore C_{13} = M_{13} = 2$

$\boxed{C_{11} = 3 \quad C_{12} = 1 \quad C_{13} = 2}$

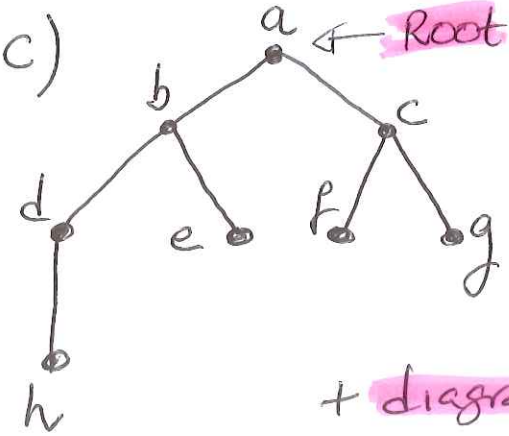
ii) determinant of $A \Rightarrow \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

$\det A = (1)(3) + (-1)(1) + (2)(2)$

$= 3 - 1 + 4$

$\boxed{\det A = 6}$

c)



a ← Root node

node c is the parent of f and g

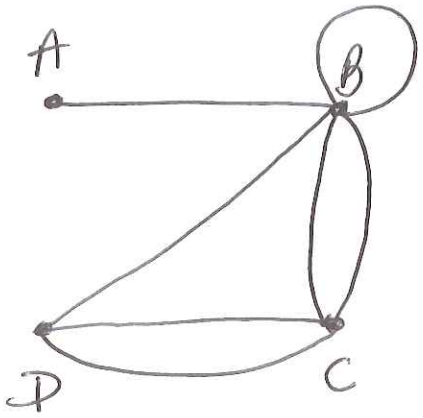
f and g are children of c

h, e, f and g are leaf nodes...

+ diagram.

d)

	a	b	c	d
a	0	1	0	0
b	1	1	2	1
c	0	2	0	2
d	0	1	2	0



$\deg A = 1,\ \deg B = 6,\ \deg C = 4\ \deg D = 3$

e) 18, 10, 27, 6, 28, 11, 2, 26.

i) Mean: use $\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n} = \dfrac{18+10+27+6+28+11+2+26}{8}$

$= \dfrac{128}{8}$

$\boxed{\bar{x} = 16}$

ii) Standard deviation: use $s^2 = \dfrac{\sum_{i=1}^{n} x_i^2 - n(\bar{x})^2}{n-1}$

$s^2 = \dfrac{18^2+10^2+27^2+6^2+28^2+11^2+2^2+26^2 - 8(16)^2}{8-1}$

$= \dfrac{324+100+729+36+784+121+4+676-2048}{7}$

$= \dfrac{2774-2048}{7} = \dfrac{726}{7}$

$s^2 = 103.7$

$s = \sqrt{103.7} = \boxed{10.18}$

f) i) $P[5] = \frac{1}{6}$

ii) $P[\text{product} < 7]$ => Build the event space...

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

$P[\text{product} < 7] = \frac{14}{36} = \frac{7}{18}$

g) $y = 3x^3 - x$ from first principles

↑ must show all work.

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$f(x) = y = 3x^3 - x$

$f(x+h) = 3(x+h)^3 - (x+h)$

$= 3(x^2 + 2xh + h^2) - (x+h)$

$= 3x^2 + 6xh + 3h^2 - x - h$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{3x^2 + 6xh + 3h^2 - x - h - (3x^2 - x)}{h}$$

$$= \lim_{h \to 0} \frac{6xh + 3h^2 - h}{h} = \lim_{h \to 0} 6x + 3h - 1$$

$$\boxed{\frac{dy}{dx} = 6x - 1}$$

h) indefinite integrals.

i) $\int \frac{6}{x}\,dx = 6\int \frac{1}{x}\,dx = 6\ln x + C$

ii) $\int (8x^3 + 8x^2 - 6x + 7)\,dx$

$= \frac{8x^4}{4} + \frac{3x^3}{3} - \frac{6x^2}{2} + 7x + C$

$= 2x^4 + x^3 - 3x^2 + 7x + C$

Question 2: Differentiation and Integration

a) slope of tangent to $y = 2x^3 - 3x + 6$ at $(2, 10)$

↓ get $\frac{dy}{dx}$

$\frac{dy}{dx} = 6x^2 - 3$ sub in $x = 2$ to get slope

$= 6(2)^2 - 3$

$\boxed{= 21}$ ← slope of the tangent at the point $(2, 10)$

b) $f(x) = \underbrace{(3x+2)}_{u}\underbrace{(5x-7)}_{v}$

Product Rule

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$u = (3x+2) \quad \frac{du}{dx} = 3$

$v = (5x-7) \quad \frac{dv}{dx} = 5$

$$\frac{df}{dx} = (3x+2)(5) + (5x-7)(3)$$
$$= 15x + 10 + 15x - 21$$
$$= 30x - 11$$

c) Max, Min points + point of inflection.

using 2nd derivative test: $y = 8x^3 + 4x^2 - 6$.

- get $\frac{dy}{dx}$
- let $\frac{dy}{dx} = 0$ and solve for x
- get $\frac{d^2y}{dx^2}$
- sub x into $\frac{d^2y}{dx^2}$

if $\frac{d^2y}{dx^2} < 0$ MAX POINT

$\frac{d^2y}{dx^2} > 0$ MIN POINT.

$\frac{dy}{dx} = 24x^2 + 8x$

factorise: $(24x+8)(x) = 0$

$24x + 8 = 0$

$x = -1/3$

or

$x = 0$.

$\frac{d^2y}{dx^2} = 48x + 8$

$x = -1/3 : 48(-1/3) + 8 = -8$ MAX

$x = 0 : 48(0) + 8 = 8$ MIN

when $x = 0$ $\quad y = 8(0)^3 + 4(0)^2 - 6$

$y = -6$

MIN point $= (0, -6)$

when $x = -\frac{1}{3}$ $\quad y = 8\left(\frac{1}{3}\right)^3 + 4\left(\frac{1}{3}\right)^2 - 6$

$= \frac{-8}{27} + \frac{4}{9} - 6$

$= \frac{-158}{27}$

MAX point $= \left(-\frac{1}{3}, -\frac{158}{27}\right)$

POINT OF INFLECTION → change in concavity.

$\frac{d^2y}{dx^2} = 48x + 8 = 0$

$48x = -8$

$x = \frac{-8}{48} = -\frac{1}{6}$.

Take a point to the left and right of $x = -\frac{1}{6}$.

let $x = -0.18$ $\quad 48(-0.18) + 8 = -0.64 < 0$ concave down

$x = -0.15$ $\quad 48(-0.15) + 8 = 0.8 > 0$ concave up.

when $x = -\frac{1}{6}$ $\quad y = 8\left(-\frac{1}{6}\right)^3 + 4\left(-\frac{1}{6}\right)^2 - 6$

$y \approx -5.93$

point of inflection $(-0.167, -5.926)$

d) $y = x + 2x^2 \qquad x = 0 \qquad x = 2$

$$\int_0^2 x + 2x^3 = \frac{x^2}{2} + \frac{2x^3}{3}\Big]_0^2$$

$$= \left[\frac{2^2}{2} + \frac{2(2)^3}{3}\right] - \left[\frac{0^2}{2} + \frac{2(0)^3}{3}\right]$$

$$= \left[\frac{4}{2} + \frac{16}{3}\right] - [0]$$

$$= \frac{44}{6} = \frac{22}{3} \text{ square units.}$$

Question 3: Statistics & Probability.

a) see notes.

b) **GROUPED DATA.**

Exam Score	f_i no. students	(i) Rel Freq	M_i (Midpoint)	$f_i M_i$	M_i^2	$f_i M_i^2$
0-9	1	1/42 = 0.02	4.5	4.5	20.25	20.25
10-19	1	1/42 = 0.02	14.5	14.5	210.25	210.25
20-29	2	2/42 = 0.05	24.5	49	600.25	1200.5
30-39	5	5/42	34.5	172.5	1190.25	5951.25
40-49	7	7/42	44.5	311.5	1980.25	13,861.75
50-59	5	5/42	54.5	272.5	2970.25	14,851.25
60-69	8	8/42	64.5	516	4160.25	33,282
70-79	6	6/42	74.5	447	5550.25	33,301.5
80-89	4	4/42	84.5	338	7140.25	28,561
90-99	3	3/42	94.5	283.5	8930.25	26,790.75
	total: 42			2409		158,030.5

ii) mean: use $\bar{x} = \dfrac{\sum_{i=1}^{n} f_i m_i}{\sum f_i} = \dfrac{2409}{42} = 57.4$.

iii) standard deviation $s^2 = \dfrac{\sum_{i=1}^{M} f_i M_i^2 - M(\bar{x})^2}{M-1} = \dfrac{158,030.5 - 42 \times (57.4)^2}{42-1}$

$$s^2 = \frac{19650.58}{41} = 479.28$$

$$s = \sqrt{479.28} = \boxed{21.9}$$

c) i) order matters ⇒ permutation

$${}^{6}P_{3} = \frac{n!}{(n-r)!} = \frac{6!}{(6-3)!} = \frac{6!}{3!} = 120$$

ii) order doesn't matter ⇒ combinations.

$${}^{6}C_{3} = \frac{n!}{(n-r)!r!} = \frac{6!}{(6-3)!3!} = \frac{6\times5\times4\times3\times2\times1}{(3\times2\times1)(3\times2\times1)}$$

$$= \frac{120}{6}$$

$$= 20$$

d) normal distribution

$$a = \frac{x-\mu}{\sigma} \qquad \mu = 1.98 \qquad \sigma = 0.012$$

$$P[X > 2] \equiv P[Z > a] \quad \text{Find } a.$$

$$a = \frac{2-1.98}{0.012} = 1.67$$

$$P[Z > 1.67] \quad \text{look up table}$$

$$\boxed{= 0.0475}$$

Question 4: Matrices, Graphs & Trees.

a) $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 5 \\ 6 & 0 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 9 & 1 \end{bmatrix}$

i) calculate AA^T.

$A^T = \begin{bmatrix} 1 & 4 & 6 \\ 2 & 1 & 0 \\ 3 & 5 & 2 \end{bmatrix}$

Transpose:
Rows become columns and columns become rows.

$AA^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 5 \\ 6 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 6 \\ 2 & 1 & 0 \\ 3 & 5 & 2 \end{bmatrix}$

remember -
Must do multiplication in the order given in the question, ...

row / column.

$= \begin{bmatrix} \text{R1xC1}: 1\times1+2\times2+3\times3 & \text{R1xC2}: 1\times4+2\times1+3\times5 & \text{R1xC3}: 1\times6+2\times0+3\times2 \\ \text{R2xC1}: 4\times1+1\times2+5\times3 & \text{R2xC2}: 4\times4+1\times1+5\times5 & \text{R2xC3}: 4\times6+1\times0+5\times2 \\ \text{R3xC1}: 6\times1+0\times2+2\times3 & \text{R3xC2}: 6\times4+0\times1+2\times5 & \text{R3xC3}: 6\times6+0\times0+2\times2 \end{bmatrix}$

$= \begin{bmatrix} 1+4+9 & 4+2+15 & 6+0+6 \\ 4+2+15 & 16+2+25 & 24+0+10 \\ 6+0+6 & 24+0+10 & 36+0+4 \end{bmatrix} = \begin{bmatrix} 14 & 21 & 12 \\ 21 & 43 & 34 \\ 12 & 34 & 40 \end{bmatrix}$

ii) inverse of B. $B = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 9 & 1 \end{bmatrix}$

Steps
- get cofactors - C and C^T
- get the determinant B
- sub into the formula for inverse $B^{-1} = \frac{1}{\det B} C^T$

Cofactors $\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$

C_{11}: cross out row 1, column 1 → submatrix $\begin{vmatrix} 3 & 1 \\ 9 & 1 \end{vmatrix}$ (a, b, c, d) get ad-bc ie. M_{11} (Minor)

$(3)(1) - (1)(9) = 3 - 9 = -6$

$C_{11} = M_{11} = \boxed{-6}$

C_{12}: cross out row 1, column 2 → submatrix $\begin{vmatrix} 2 & 1 \\ 4 & 1 \end{vmatrix}$ ad-bc $2(1) - 1(4) = -2$ ↙ M_{12}

$i+j = 1+2 = (3)$ odd $\therefore C_{12} = -M_{12} = -(-2) = \boxed{2}$

C_{13}: cross out row 1, column 3 → submatrix $\begin{vmatrix} 2 & 3 \\ 4 & 9 \end{vmatrix}$ ad-bc $2(9) - 3(4) = 6$ ↙ M_{13}

$i+j = 1+3 = (4)$ even $\therefore C_{13} = M_{13} = \boxed{6}$

over →

C_{21} : $\begin{vmatrix} 1 & 1 \\ 2 & 3 \\ 4 & 9 \end{vmatrix}$ ~~1~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~9~~ 1 1

submatrix $\begin{vmatrix} 1 & 1 \\ 9 & 1 \end{vmatrix}$ ad-bc $1(1)-1(9) = -8$ ← M_{21}

$i+j = 2+1 = (3)^{odd}$ ∴ $C_{21} = -M_{21} = -(-8) = \boxed{8}$

C_{22} : 1 1 ~~2~~ ~~3~~ ~~4~~ ~~9~~ 1 1

submatrix $\begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix}$ ad-bc $1(1)-1(4) = -3$ ← M_{22}

$i+j = 2+2 = (4)^{even}$ ∴ $C_{22} = M_{22} = \boxed{-3}$

C_{23} : 1 1 ~~2~~ ~~3~~ 1 4 9

Submatrix $\begin{vmatrix} 1 & 1 \\ 4 & 9 \end{vmatrix}$ ad-bc $1(9)-1(4) = 5$ ← M_{23}

$2+3 = (5)^{odd}$

∴ $C_{23} = -M_{23} = -(5) = \boxed{-5}$

C_{31} : 1 1 1 2 3 1 ~~4~~ ~~9~~ ~~1~~

submatrix $\begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix}$ ad-bc $1-3 = -2$ ← M_{31}

$3+1 = (4)^{even}$

∴ $C_{31} = M_{31} = \boxed{-2}$

over ⟶

C_{32} : 1 1 / 2 3 1 / 4 9 1

$\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}$ ad-bc

$1 - 2 = -1$ ← M_{32} $3+2=5$ odd

$\therefore C_{32} = -M_{32} = -(-1) = \boxed{1}$

C_{33} : 1 1 / 2 3 / 4 9

$\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$ ad-bc

$3 - 2 = 1$ ← M_{33} $3+3=6$ even

$\therefore C_{33} = M_{33} = \boxed{1}$

Matrix of cofactors $C = \begin{bmatrix} -6 & 2 & 6 \\ 8 & -3 & -5 \\ -2 & 1 & 1 \end{bmatrix}$

Determinant $\det B = b_{11}C_{11} + b_{12}C_{12} + b_{13}C_{13}$

$= (1)(-6) + (1)(2) + (1)(6)$

$= -6 + 2 + 6$

$\boxed{\det B = 2}$

over ⟶

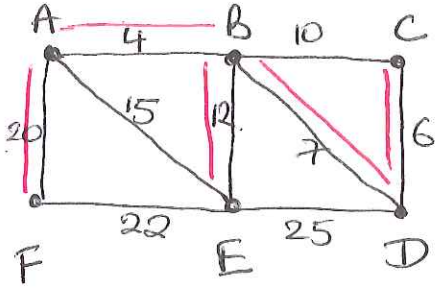
Inverse $B^{-1} = \frac{1}{\det B} C^T$

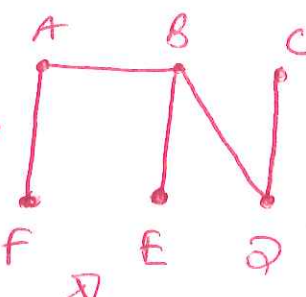
$$C^T = \begin{bmatrix} -6 & 8 & -2 \\ 2 & -3 & 1 \\ 6 & -5 & 1 \end{bmatrix}$$

$$\Rightarrow B^{-1} = \frac{1}{2} \begin{bmatrix} -6 & 8 & -2 \\ 2 & -3 & 1 \\ 6 & -5 & 1 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} -3 & 4 & -1 \\ 1 & -\frac{3}{2} & \frac{1}{2} \\ 3 & -\frac{5}{2} & \frac{1}{2} \end{bmatrix}$$

b) MST - Minimal spanning tree





Answer

$T = \{A\,ab\,B\,bd\,D\,dc\,C\,be\,E\,af\,F\}$

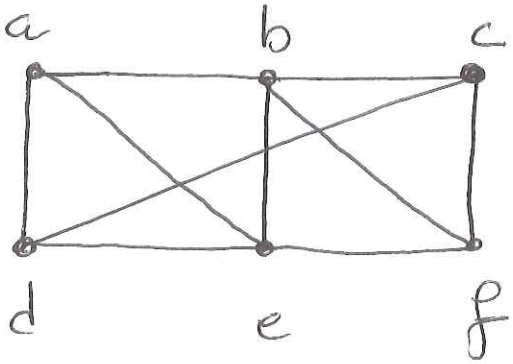
MST

Prim's Algorithm

- Start at A, pick edge with the least weight (incident to A) i.e. A → B weight 4.
- Look for edge with least weight incident to A and B. → BD weight 7
- Look for edge with least weight incident to A, B and D → DC weight 6
- Look for edge with least weight incident to A, B, D and C → BC, however this causes a circuit i.e. B → D → C - B. Therefore select the edge with the next least weight → BE weight 12
- Look for edge with least weight incident to A, B, D, C and E → AF weight 20

Finished N nodes, N-1 edges.

c) paths of length 2 between a and f.



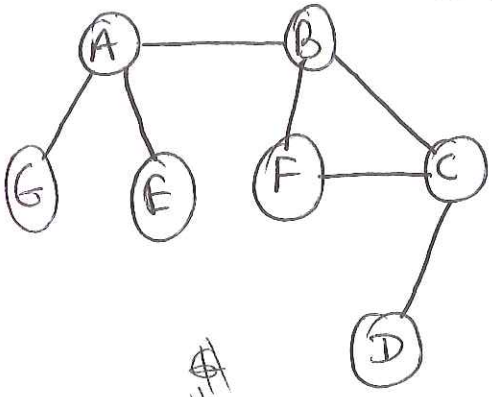
paths of length 2
- a → b → f.
- a → e → f

d) difference between BFS and DFS, use examples

Depth-first and Breadth-first traversals are algorithms for searching or traversing graph or tree structures.

DFS → uses a stack data structure to complete traversal.
- starts at a given node (OR root node if a tree structure) and traverses/moves as far as possible away/along each edge before backtracking.

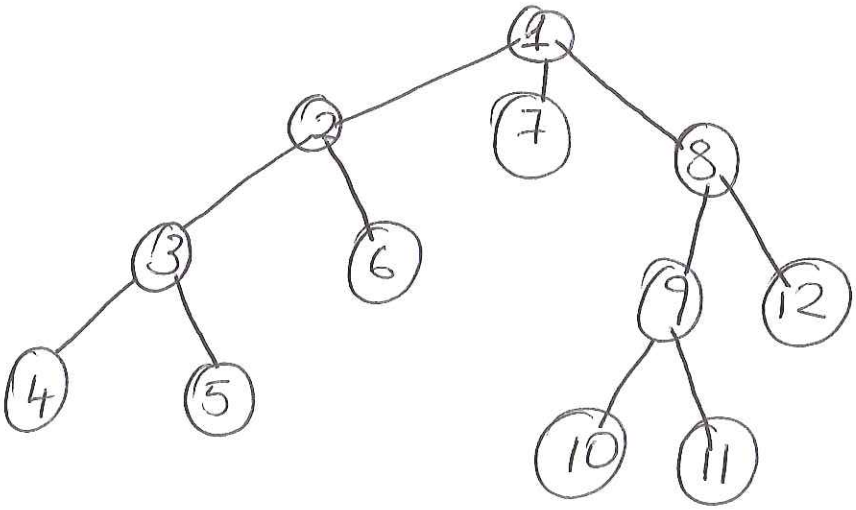
e.g. Graph G start at A
choose alphabetically.



~~A~~ ~~B~~ ~~F~~ ~~D~~ ~~C~~ ~~G~~ ~~E~~
B
Stack: A

Output: A B C D F E G

Tree structure.

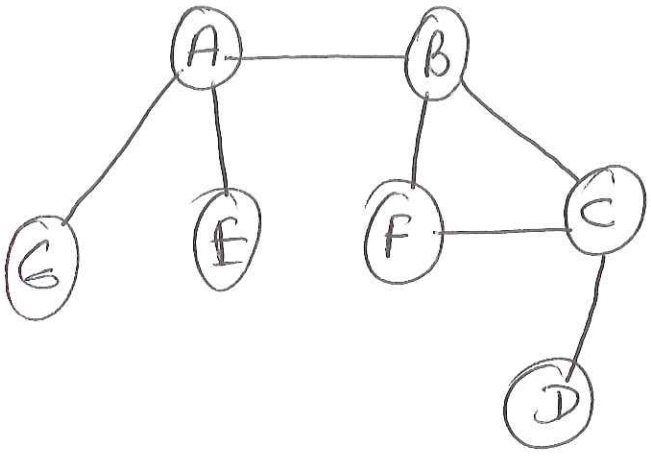


numbers 1-12, order in which the tree nodes are traversed/visited.

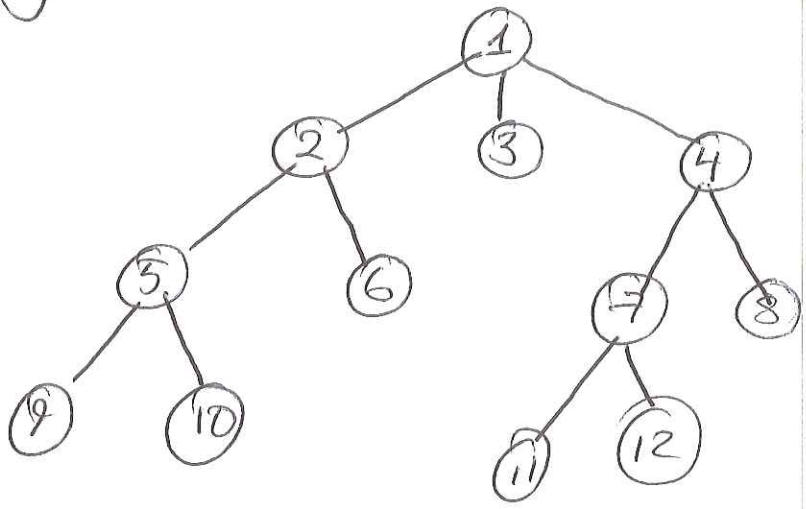
Breadth-first traversal → starts at given node (eg. node A) or root node, if a tree structure.
uses a Queues data structure → traverses adjacent/neighbour nodes (layer by layer).

↓

e.g. Graph G • start at A
• choose alphabetically



Rooted TREE



Queue: B
E
G
C
D
F

numbers 1-12, order in which the nodes are visited and output.

output: ABEGCFD